U.G.C. NET Exam.
COMPUTE
(Solved with

1. Which of the statements given below is/are correct?

It is always important and useful to include an 'alt' attribute on 'img' tag in HTML because

A. users who cannot see the image due to vision impairment can have a textual description of the image (which can be spoken aloud by a screen reader).

B. If the image fails to load (slow connection, broken path, etc.) then alt text is displayed instead.

C. SEO (Search Engine Optimization) benefits.

Choose the correct answer from the options given below:

(a) A only
(b) B and C only
(c) C only
(d) A, B, and C

Ans. (d) : When an image contains words that are important to understanding the context, the all text should include those words. This will allow the alt text to play the same junction on the page as the image. Note that it does not necessarily describe the visual characteristics of the image itself but must convey the same meaning as the image.
Example:- An image on a website provides a link to a free newsletter. The image contains the text "Free newsletter. Get free recipes, news, and more. Learn more". the alt text matches the text in the image.

2. What does the following function f() in 'C' return?

int f(unsigned int N) { unsigned int counter = 0; while(N > 0) { counter += N & 1;
N = N >> 1;} return counter == 1;}

(a) 1 if N is odd, otherwise 0
(b) 1 if N is a power of 2, otherwise 0
(c) 1 if the binary representation of N is all 1's, otherwise 0
(d) 1 if the binary representation of N has any 1's, otherwise 0

Ans. (b) : 1 if N is a power of 2, otherwise 0

3. Consider the following recursive function F() in Java that takes an integer value and returns a string value:

public static String F(int N) {if (N <= 0) return "–"; return F(N - 3) + N + F(N - 2) + N;}

The value of F(5) is:

(a) -2-25-3-135
(b) -2-25-1-3-135
(c) -1-145-2-245
(d) -2-25-3-1-135

Ans. (d) : -2-25-3-1-135

4. Match List-I with List-II

	List - I (Programming Term)		List - II (Meaning)
A.	**JNDI**	**i.**	**Runtime support for running Java programs**
B.	**RMI**	**ii.**	**The API in support of naming and directory services**
C.	**JDK**	**iii.**	**The methods provided by the Java development kit and runtime support for calling remote methods**
D.	**JRE**	**iv.**	**The compiler and class libraries to develop Java applications**

Choose the correct answer from the options given below:
(a) A - II , B - III, C - IV, D - I
(b) A - II, B - IV, C - III, D - I
(c) A - I, B - III, C - IV, D - II
(d) A - III, B - II, C - IV, D - I

Ans. (a) :

	List - I (Programming Term)		List - II (Meaning)
A.	JNDI	ii.	The API in support of naming and directory services
B.	RMI	iii.	The methods provided by the Java development kit and runtime support for calling remote methods

C.	JDK	iv.	The compiler and class libraries to develop Java applications
D.	JRE	i.	Runtime support for running Java programs

5. Match List-I with List-II

	List - I (Programming Paradigm)		**List - II (Characteristic)**
A.	**Imperative**	**i.**	**Declarative, clausal representation, theorem proving**
B.	**Object-oriented**	**ii.**	**Side-effect free, declarative, expression evaluation**
C.	**Logic**	**iii.**	**Imperative, abstract data type**
D.	**Functional**	**iv.**	**Command-based, procedural**

Choose the correct answer from the options given below:

(a) A - IV, B - III, C - I, D - II
(b) A - III, B - IV, C - I, D - II
(c) A - IV, B - III, C - II, D - I
(d) A - II, B - III, C - I, D - IV

Ans. (a) :

	List - I (Programming Paradigm)		**List - II (Characteristic)**
A.	Imperative	iv.	Command-based, procedural
B.	Object-oriented	iii.	Imperative, abstract data type
C.	Logic	i.	Declarative, clausal representation, theorem proving
D.	Functional	ii.	Side-effect free, declarative, expression evaluation

6. Suppose you have eight 'black and white' images taken with a 1-megapixel camera and one '8-color' image taken by an 8-megapixel camera. How much hard disk space in total do you need to store these images on your computer?

(a) 1 GB (b) 4 MB
(c) 3 MB (d) 3 GB

Ans. (b) : 4 MB

Cam-1

8 block and white image {1 bit}

1-megapixel = 10^6

$\Rightarrow \quad 8 \times 1 \text{ bit} = 8\times\frac{1}{8} \text{ B} = 1 \text{ byte}$

$\Rightarrow \quad 1 \text{ B} \times 1 \times 10^6 = 1\times10^6 \text{ B} = 1 \text{ MB}$

Cam-2

1 8-color image {3 bit}

8-magapixel

$= 1 \times 3 \text{ bit} \Rightarrow 1\times\frac{3}{8} \text{ byte}$

$\Rightarrow \quad 8\times10^6 \times\frac{3}{8} \text{ byte}$

$= 3\times10^6$ Byte

= 3 MB

Total = 1 MB + 3MB = 4MB

7. Which of the statements given below are correct?

The midpoint (or Bresenham) algorithm for rasterizing lines is optimized relative to DDA algorithm in that

A. it avoids round-off operations.

B. it is incremental.

C. it uses only integer arithmetic.

D. all straight lines can be displayed as straight (exact).

Choose the correct answer from the options given below:

(a) A and B only
(b) A and C only
(c) A, B, C, and D
(d) A, B, and C only

Ans. (d) : A, B, and C only

8. What is the transformation matrix M that transforms a square in the xy-plane defined by $(1, 1)^T$, $(-1, 1)^T$, $(-1, -1)^T$ and $(1, -1)^T$ to a parallelogram whose corresponding vertices are $(2, 1)^T$, $(0, 1)^T$, $(-2, -1)^T$ and , $(0, -1)^T$?

(a) $M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(b) $M = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Ans. (a) : $M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

9. Suppose a Bezier curve p(t) is defined by the following four control points in the xy-plane: $P_0 = (-2,0); P_1 = (-2,4); P_2 = (2,4)$; and P_3 = (2,0). Then which of the following statements are correct?

A. Bezier curve P(t) has degree 3.

B. $P\left(\frac{1}{2}\right) = (0, 3)$.

C. Bezier curve P(t) may extend outside the convex hull of its control points.

Choose the correct answer from the options given below :

(a) A and B only
(b) A and C only
(c) B and C only
(d) A, B, and C

Ans. (a) : A and B only

10. Given below are two statements

Statement I: The maximum number of sides that a triangle might have when clipped to a rectangular viewport is 6.

Statement II: In 3D graphics, the perspective transformation is nonlinear in z.

In light of the above statements, choose the correct answer from the options given below

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (d) : Statement I is false but Statement II is true

11. In software testing, beta testing is the testing performed by _______________.

(a) potential customers at the developer's location
(b) potential customers at their own locations
(c) product developers at the customer's location
(d) product developers at their own locations

Ans. (b) : Beta testing is a type of user acceptance testing among the most crucial testing, which performed before the release of the software. Beta testing is a type of field test. This testing performs at the end of the software testing life cycle. This type of testing can be considered as external user acceptance testing.

Beta testing always is done after the alpha testing, and before releasing it into the market. Beta testing is a black-box testing Beta testing performs in the absence of tester and the presence of real users. Beta testing generally is done for testing software product like utilities, operating systems, and applications, etc.

12. The V components in MVC are responsible for:

(a) User interface.

(b) Security of the system.

(c) Business logic and domain objects.

(d) Translating between user interface actions/events and operations on the domain objects.

Ans. (a) : In model-view-controller architectural pattern, there are following parts and there main responsibilities-

Model: Responsible for providing the app's "business logic".

View: Responsible for providing the app's user interface (UI)

Controller: Responsible for translating UI actions to operations on the model.

So V components in MVC are responsible for user interface.

13. In software engineering, what kind of notation do formal methods predominantly use?

(a) textual

(b) diagrammatic

(c) mathematical

(d) computer code

Ans. (c) : mathematical

Formal method :- System design techniques that use rigorously specified mathematical model to build software, hardware system.

14. If every requirement stated in the Software Requirement Specification (SRS) has only one interpretation, then SRS is said to be

(a) correct

(b) consistent

(c) unambiguous

(d) verifiable

Ans. (c) : An SRS is said to be unambiguous if all the requirements stated have only 1 interpretation.

15. A system has 99.99% uptime and has a mean-time-between-failure of 1 day. How fast does the system has to repair itself in order to reach this availability goal?

(a) ~9 Seconds

(b) ~10 Seconds

(c) ~11 Seconds

(d) ~12 Seconds

Ans. (a) : ~9 Seconds

Sol.

$$\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

= MTBF (in second)

= 1 day

= 1×24×60×60

= 86400 sec

$\Rightarrow \quad 99.99\% = \frac{86400}{86400+x}$

$$86400 + x = \frac{86400}{0.9999} \times 10000$$

86400 + x = 86408.6

x = 86408.6 – 86400

$\Rightarrow \quad x = 8.6$

$\Rightarrow \quad x = 9$ sec

16. In the context of Software Configuration Management (SCM), what kind of files should be committed to your source control repository?

A. Code files

B. Documentation files

C. Output files

D. Automatically generated files that are required for your system to be used

Choose the correct answer from the options given below:

(a) A and B only
(b) B and C only
(c) C and D only
(d) D and A only

Ans. (a) : Software configuration management - software configuration management (SCM or S/WCM) is the task of tracking and controlling changes in the software, part of the larger cross-disciplinary field of configuration management. SCM practices include revision control and the establishment of baseline.

So following type of file should be committed to source control repository - A) code files B) Documentation files.

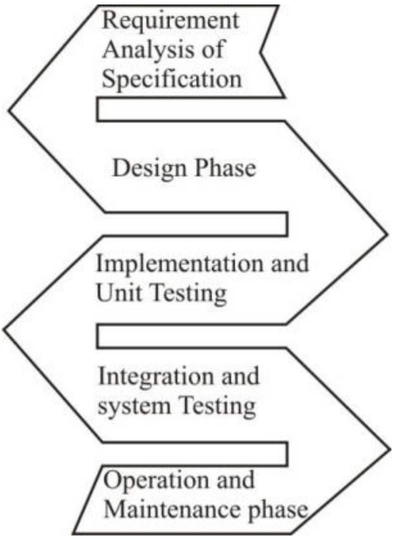
17. Match List-I with List-II

	List - I (Software Process Model)		**List - II (Description)**
A.	**Waterfall Model**	**i.**	**Software can be developed incrementally**
B.	**Evolutionary Model**	**ii.**	**Requirement compromises are inevitable**
C.	**Component - based Software Engineering**	**iii.**	**Explicit recognition of risk**
D.	**Spiral Development**	**iv.**	**Inflexible partitioning of the project into stages**

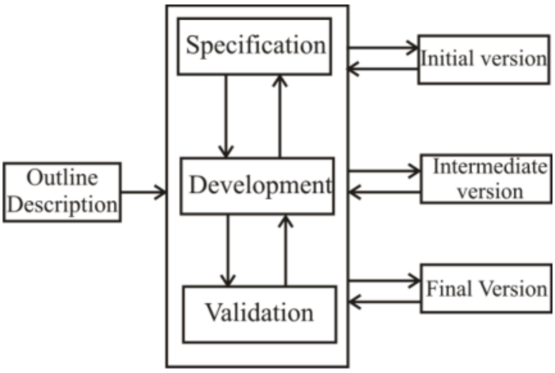
Choose the correct answer from the options given below:

(a) A - IV, B - I, C - III, D - II
(b) A - I, B - IV, C - II, D - III
(c) A - II, B - III, C - I, D - IV
(d) A - IV, B - I, C - II, D - III

Ans. (d) : Waterfall Model :-This model has five phases: Requirements analysis and specification, design, implementation and unit testing, integration and system testing and operation and maintenance. The steps always follow in this order and do not overlap. The developer must complete every phase before the next phase begins.



Evolutionary Model :- The evolutionary model is an iterative model as they are characterized in a manner that enable software engineering to develop a complete version of the software. Such models are applied because the requirement often change. So the end product will be unrealistic for a complete version is impossible due to tight market deadlines. It is better to introduce a limited version. Thus, software engineering can follow a process model that has been explicit designed to accommodate a product that gradually complete over time.



Component Based Development :- Component - based software engineering (CBSE) is an approach to software development that relies on the reuse of entities called 'software components'.

- It emerged from the failure of object - oriented development to support effective reuse single object classes are too detailed and specific.
- Components are more abstract than object classes and can be considered to be stand alone service providers. They can exist as stand-alone entities.

Spiral Development:- This software development Life Cycle models, which provides support for risk Handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project.

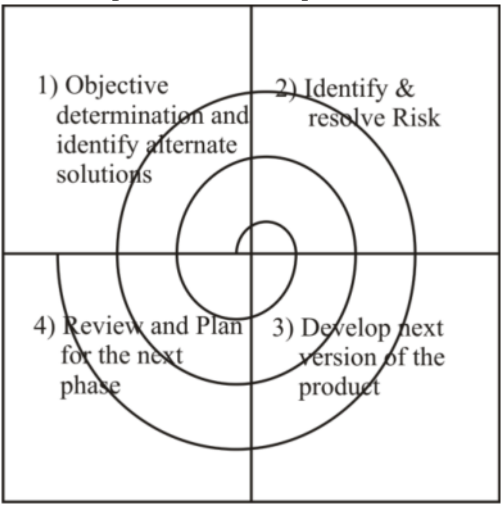
Four quadrants are

1) Objective determination and identify alternative solutions

2) Identify and resolve Risks

3) Develop next version of the product.

4) Review and plant for the next phase.



18. Identify the correct order of the following five levels of Capability Maturity Model (from lower to higher) to measure the maturity of an organization's software process.

A. Defined **B. Optimizing**

C. Initial **D. Managed**

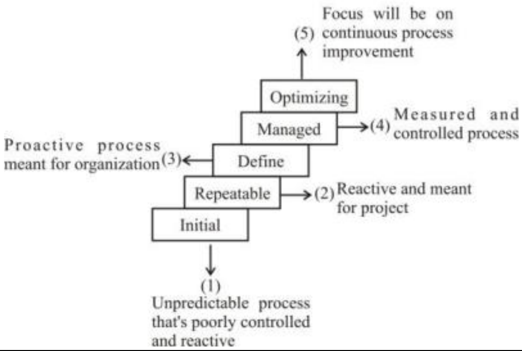
E. Repeatable

Choose the correct answer from the options given below

(a) C, A, E, D, B (b) C, E, A, B, D

(c) C, B, D, E, A (d) C, E, A, D, B

Ans. (d) :



19. Given below are two statements

Statement I: Cleanroom software process model incorporates the statistical quality certification of code increments as they accumulate into a system.

Statement II: Cleanroom software engineering follows the classic analysis, design, code, test, and debug cycle to software development and focussing on defect removal rather than defect prevention.

In light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (c) : The cleanroom software engineering process is a software development process intended to produce software with a certificate level of reliability. The focus of the cleanroom process is on defect prevention, rather than defect removal.

The cleanroom approach to software development is based on five key strategies: formal specification, Incremental development, structured programming, static verification, and statistical testing of the system.

20. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Software developers donot do exhaustive software testing in practice.

Reason R : Even for small inputs, exhaustive testing is too computationally intensive (e.g., takes too long) to run all the tests.

In light of the above statements, choose the correct answer from the options given below

(a) Both **A** and **R** are true and **R** is the correct explanation of **A**
(b) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**
(c) **A** is true but **R** is false
(d) **A** is false but **R** is true

Ans. (a) : Both A and R are true and R is the correct explanation of A.

21. Which of the following are logically equivalent?

A. $\neg p \rightarrow (q - r)$ and $q \rightarrow (p \vee r)$

B. $(p \rightarrow q) \rightarrow r$ and $p \rightarrow (q \rightarrow r)$

C. $(p \rightarrow q) \rightarrow (r \rightarrow s)$ and $(p \rightarrow r) \rightarrow (q \rightarrow s)$

Choose the correct answer from the options given below :

(a) A only
(b) A and B only
(c) B and C only
(d) A and C only

Ans. (a) : A only

22. Which of these statements about the floor and ceiling functions are correct?

Statement I : $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + (1/2) \rfloor$ for all real number x

Statement II : $\lceil x + y \rceil = \lceil x \rceil + \lceil y \rceil$ for all real number's x and y

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (c) : Statement I is true but Statement II is false
Sol.
Statement I
$\Rightarrow \quad \lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + ½ \rfloor$
$\Rightarrow \quad \lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 0.5 \rfloor$
$x = 1 \quad \lfloor 2 \times 1 \rfloor = \lfloor 1 \rfloor + \lfloor 1 + 0.5 \rfloor = \lfloor 1.5 \rfloor = 1$
$2 = 1 + 1 = 2$ True
$x = 2.75 \quad \lfloor 2 \times 2.7 \rfloor = \lfloor 2.75 \rfloor + \lfloor 2.75 + 0.5 \rfloor$
$\lfloor 5.5. \rfloor = 2 + \lfloor 3.75 \rfloor = 5 = 2 + 3 =$ True
Statement -II
$\lceil x \rceil + \lceil y \rceil = \lceil x + y \rceil$
$\Rightarrow \quad \lceil 0.5 \rceil + \lceil -0.75 \rceil = \lceil 0.5 + (-0.75) \rceil$
$1 + 0 = \lceil -0.25 \rceil$
$1 \neq -0 = 0$
$1 \neq 0$ False
Statement I is true but Statement II is false.

23. How many ways are there to assign 5 different jobs to 4 different employees if every employee is assigned at least 1 job?
(a) 1024
(b) 625
(c) 240
(d) 20

Ans. (c) : 240

24. A company stores products in a warehouse. Storage bins in this warehouse are specified by their aisle, location in the aisle, and self. There are 50 aisles, 85 horizontal locations in each aisle, and 5 shelves throughout the warehouse. What is the least number of products the company can have so that at least two products must be stored in the same bin?
(a) 251
(b) 426
(c) 4251
(d) 21251

Ans. (d) : 21251
total product
$= 50 \times 85 \times 5$
$= 21250$
= 1 product in 1 bin
$= 21250 + 1$
$= 21251$

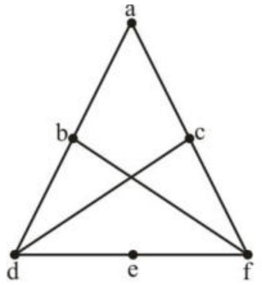
25. Let us assume a person climbing the stairs can take one stair or two stairs at a time. How many ways can this person climb a flight of eight stairs
(a) 21
(b) 24
(c) 31
(d) 34

Ans. (d) : 34

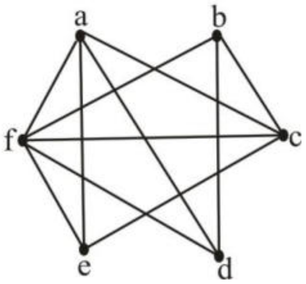
26. For which value of n is Wheel graph W_n regular?
(a) 2
(b) 3
(c) 4
(d) 5

Ans. (*) : According to UGC two option (b & c) are correct.

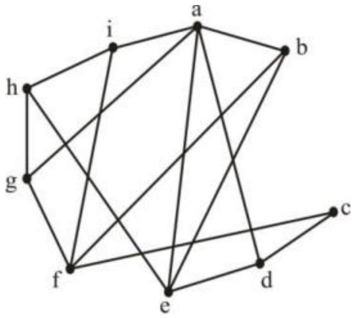
27. Which of the following Graphs is (are) planer?



(A)



(B)



(C)

Choose the correct answer from the option given below :

(a) A and B only
(b) B and C only
(c) A only
(d) B only

Ans. (a) : A planar graph is a graph that can be embedded in the plane, i.e., it can be drawn on the plane in such a way that its edges intersect only at their end points.
In other words, it can be drawn in such a way that no edges cross each other.
For a simple, connected, planar graph with v vertices and e edges and f faces, the following simple conditions hold for $v \geq 3$:
• Theorem 1. $e \leq 3v - 6$,
• Theorem 2. If there are no cycles of length 3, then $e \leq 2v - 4$
• Theorem 3. $f \leq 2v - 4$.

28. Match List-I with List-II

	List - I Identity		List - II Name
A.	**x + x =x**	**i.**	**Identity Law**
B.	**x + 0 = x**	**ii.**	**Absorption Law**
C.	**x +1 = 1**	**iii.**	**Idempotent law**
D.	**x + xy = x**	**iv.**	**Domination Law**

Choose the correct answer from the options given below:

(a) A - III, B - I, C - II, D - IV
(b) A - I, B - III, C - IV, D - II
(c) A - III, B - I, C - IV, D - II
(d) A - III, B - IV, C - I, D - II

Ans. (c) : Identify Law :- The law of identity states that each thing is identical with itself.

$x + 0 = x$

Absorption Law :- To affirm that either some proposition is true or else that proposition and some other proposition are both true is equivalent to affirming the first proposition.

$x + xy = x$

Idempotent Law :- It is the property of certain operations in mathematics and computer science that they can be applied multiple time without changing the result beyond the initial application.

$x + x = x$

$1 + 1 = 1$ and $0 \times 0 = 0$

Domination Law :- $F \wedge X = F$; if the first thing in an AND expression is false, then it doesn't matter whether the second thing is true or false. The AND expression is going to be false either way.

$T \vee X = T$: If the first thing in an OR expression is true, then, it doesn't matter whether the second thing is true or false the OR expression is going to be true either way.

29. Let (X, *) be a semigroup. Furthermore, for every a and b in X, if $a \neq b$, then $a*b \neq b*a$.

Based on the defined semigroup, choose the correct equalities from the options given below:

A. For every a in X, a*a = a

B. For every a, b in X, a*b *a= a

C. For every a, b, c in X, a*b *c= a*c

(a) A and B only
(b) A and C only
(c) B and C only
(d) A, B and C

Ans. (d) : a. Let $a*a = b.(a*a)*a = b*a$

Since (X, *) is a semi group, * is closed and associative. So,

$(a*a)*a = a*(a*a) \Rightarrow a*b = b*a$

Which is possible only if $a = b$. Thus we proved $a*a = a$

b. Let $(a*b) * a = c$

$\Rightarrow (a * b) *a *a = c * a$

$\Rightarrow a*b*a = c*a$

$\Rightarrow c * a = a$

Similarly,

$a * (a*b*a) = a*c$

$\Rightarrow a * a * (b * a) = a * c$

$\Rightarrow a * (b * a) = a * c$

$\Rightarrow a * c = a = c * a$

So, $c = a$

c. Let $(a * b)*c = d$

$\Rightarrow (a * b) * c * c = d * c$
$\Rightarrow a * b * c = d * c$
$\Rightarrow d * c = d$
Similarly
$a*(a*b*c) = a * d$
$\Rightarrow a * a *(b * c) = a * d$
$\Rightarrow a * (b * c) = a * d$
$\Rightarrow a * d = d$
Thus $d * c = a * d = d$
Now,
$c * d * c = c * a * d = c * d$
$\Rightarrow \quad c = c * a * d = c * d$
and
$d * c * a = a * d * a = d * a$
$\Rightarrow \quad d * c * a = a = d * a$
So,
$= a * c = (d * a) * (c * d)$
$= d * (a * c) * d = d$
Thus $a * b * c = a * c$

30. Consider the following linear optimization problem:
Maximize Z = 6x+5y
Subject to 2x – 3y <= 5
x+3y <= 11
4x + y <=15
and x>=0, y >= 0.
The optimal solution of the problem is:
(a) 15
(b) 25
(c) 31.72
(d) 41.44

Ans. (c) : 31.72

31. Which of the following languages are not regular?
A. L={ $(01)^n 0^k$ | n > k, k>=0 }
B. L={ $c^n b^k a^{n+k}$ | n >= 0, k>=0 }
C. L={ $0^n 1^k$ | $n \neq k$ }
Choose the *correct* answer from the options given below:
(a) A and B only
(b) A and C only
(c) B and C only
(d) A, B and C

Ans. (d) : A, B and C , no one are regular

32. Any string of terminals that can be generated by the following context free grammar (where S is start non terminal symbol)
$S \rightarrow XY$
$X \rightarrow 0X \mid 1X \mid 0$
$Y \rightarrow Y0 \mid Y1 \mid 0$

(a) has at least one 1
(b) should end with 0
(c) has no consecutive 0's or 1's
(d) has at least two 0's

Ans. (d) : A language, which has at least two 0's

33. Let

$L_1 = \{ 0^n1^n0^m \mid n>=1, m>=1 \}$
$L_2 = \{ 0^n 1^m 0^m \mid n>=1, m>=1 \}$
$L_3 = \{ 0^n 1^n 0^n \mid n>=1\}$

Which of the following are correct statements?

A. $L_3 = L_1 \cap L_2$
B. L_1 and L_2 are context free languages but L_3 is not a context free language
C. L_1 and L_2 are not context free languages but L_3 is a context free language
D. L_1 is a subset of L_3

Choose the correct answer from the options given below:

(a) A and B only
(b) A and C only
(c) A and D only
(d) A only

Ans. (a) : A and B are correct option.

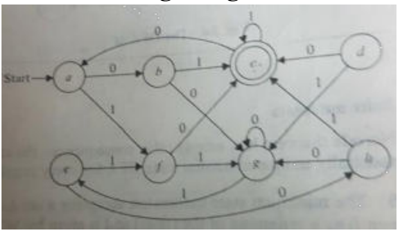
34. Given below are two statements

Statement I: The family of context free languages is closed under homomorphism
Statement II: The family of context free languages is closed under reversal
In light of the above statements, choose the correct answer from the options given below :

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (a) : Both Statement I and Statement II are true

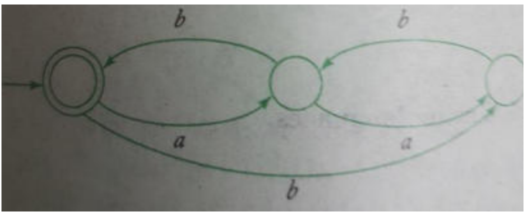
35. What is the minimum number of states required to the finite automaton equivalent to the transition diagram given below?



(a) 3
(b) 4
(c) 5
(d) 6

Ans. (c) : 5

36. Find the regular expression for the language accepted by the automata given below.



(a) (aa* (a+b)ab)*
(b) (a+b)ab(ab+bb+ aa* (a+b)ab)*
(c) a* ab(ab+bb+ aa* (a+b)ab)*
(d) a* (a+b)ab(ab+bb+ aa* (a+b)ab)*

Ans. (*) :

37. What language is accepted by the pushdown automaton
$M=(\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, z\}, \delta, q_0, z, \{q\})$
with $\delta(q_0, a, a) = \{(q_0, aa)\}$; $\delta(q_0, b, a) = \{(q_0, ba)\}$
$\delta(q_0, a, b) = \{(q_0, ab)\}$; $\delta(q_0, b, b) = \{(q_0, bb)\}$
$\delta(q_0, a, z) = \{(q_0, az)\}$; $\delta(q_0, b, z) = \{(q_0, bz)\}$
$\delta(q_0, \lambda, b) = \{(q_1, b)\}$; $\delta(q_0, \lambda, a) = \{(q_1, a)\}$
$\delta(q_1, a, a) = \{(q_1, \lambda)\}$; $\delta(q_1, b, b) = \{(q_1, \lambda)\}$
$\delta(q_1, \lambda, z) = \{(q_2, z)\}$?

(a) $L = \{ w \mid n_a(w) = n_b(w), w \in \{a, b\}^+ \}$
(b) $L = \{ w \mid n_a(w) <= n_b(w), w \in \{a, b\}^+ \}$
(c) $L = \{ w \mid n_b(w) <= n_a(w), w \in \{a, b\}^+ \}$
(d) $L = \{ww^R \mid w \in \{a, b\}^+ \}$

Ans. (d) : $L = \{ww^R \mid w \in \{a, b\}^+ \}$

38. Match List - I with List - II

	List - I Production Rules		**List - II Grammar**
A.	**$S \to XY$ $X \to 0$ $Y \to 1$**	**I.**	**Greibach Normal Form**
B.	**$S \to aS \mid bSS \mid c$**	**II.**	**Context Sensitive Grammar**
C.	**$S \to AB$ $A \to 0A \mid 1A \mid 0$ $B \to 0A$**	**III.**	**Chomsky Normal Form**
D.	**$S \to aAbc$ $Ab \to bA$ $Ac \to Bbcc$ $bB \to Bb$ $aB \to aa \mid aaA$**	**IV.**	**S - Grammar**

Choose the correct answer from the options given below:

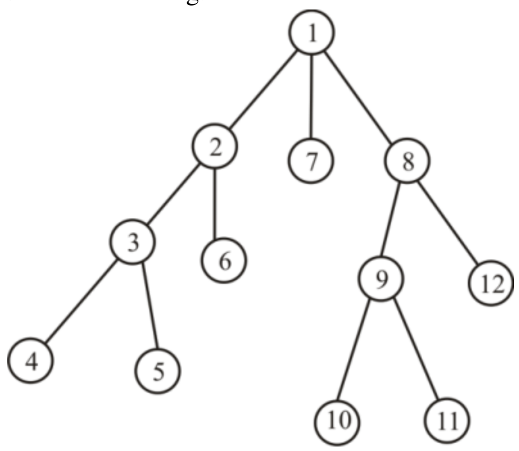
(a) A - III, B - I , C - IV, D - II
(b) A - III, B - II , C - I, D - IV
(c) A - III, B - IV, C - I, D - II
(d) A - IV, B - III , C - I, D - II

Ans. (*) :

39. Which of the following concepts can be used to identify loops?
A. Depth first ordering
B. Dominators
C. Reducible graphs
Choose the correct answer from the options given below:
(a) A and B only
(b) A and C only
(c) B and C only
(d) A, B and C

Ans. (d) : A Depth first ordering :- Depth - first search (DFS) is an algorithm for traversing or searching tree or graph data structures. The algorithm starts at the root node (selecting some orbitrary node as the root node in the case of a graph) and explores as for as possible along each branch before back tracking.

Vertex Orderings :-
i) pre ordering
ii) post ordering
iii) reverse preordering
iv) reverse postordering
for binary trees there is additionally-
e) in ordering
f) reverse in - ordering.



Dominators :- Dominators are defined in a directed graph with respect to a source, vertex S. formally, a node a is said to dominate a node w art source vertex S if all the paths from S to w in the graph must pass through node u. Note that only the vertices that are reachable from source vertex in the directed graph are considered here.

Reducible Graph :- A control how graph G is said to be reducible if the removal of its back edges leads to an acyclic graph where each node can be reached from the initial node of G.

So All above concepts can be used to identify loops.

40. Given below are two statements
Statement I: LL(1) and LR are examples of Bottom - up parsers.
Statement II: Recursive descent parser and SLR are examples of Top - down parsers
In light of the above statements, choose the correct answer from the options given below:
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (b) : Both Statement I and Statement II are false.

41. The postfix form of the expression (A + B) * (C * D - E) * F / G is _______ .
(a) A B + C D * E − F G / * *
(b) A B + C D * E − F * * G /
(c) A B + C D * E − * F * G /
(d) A B + C D E * − * F * G /

Ans. (c) : A B + C D * E − * F*G/
Sol. postfix expression-
⇒ ((A + B) *((C + D) – E)) * F)/G
⇒ ((AB +) * ((CD *) E –)) F *) G/
⇒ ((AB +) (CD * E –) *)) F *) G /
⇒ **AB + CD * E – * F * G /**

42. A double - ended queue (dequeue) supports adding and removing items from both the ends of the queue. The operations supported by dequeue are AddFront(adding item to front of the queue), AddRear(adding item to the rear of the queue), RemoveFront(removing item from the front of the queue), and RemoveRear(removing item from the rear of the queue). You are given only stacks to implement this data structure. You can implement only push and pop operations. What's the time complexity of performing AddFront() and AddRear() assuming *m* is the size of the stack and *n* is the number of elements?
(a) O(m) and O(n)
(b) O(1) and O(n)
(c) O(n) and O(1)
(d) O(n) and O(m)

Ans. (b) : O(1) and O(n)

43. Two balanced binary trees are given with m and n elements, respectively. They can be merged into a balanced binary search tree in ____ time.
(a) O(m*n)
(b) O(m+n)
(c) O(m*log n)
(d) O(m*log(m+n))

Ans. (b) : First we store the inorder traversals of both the trees in two separate arrays and then we can merge these sorted sequences in O(m+n) time. And then we construct the balanced tree from this final sorted array.

44. **A data structure is required for storing a set of integers such that each of the following operations can be done in O(log n) time, where *n* is the number of elements in the set.**

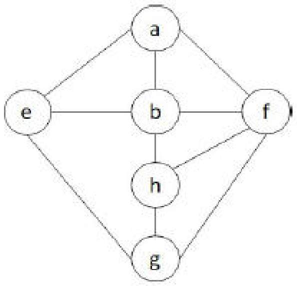
- **Deletion of the smallest element**
- **Insertion of an element if it is not already present in the set**

Which of the following data structures can be used for this purpose?

(a) A heap can be used but not a balanced binary search tree.
(b) A balanced binary search tree can be used but not a heap.
(c) Both balanced binary search tree and heap can be used.
(d) Neither balanced binary search tree nor heap can be used.

Ans. (b) : A balanced binary search tree can be used but not a heap.

45. **Consider the following graph.**



Among the following sequences
I. a b e g h f
II. a b f e h g
III. a b f h g e
IV. a f g h b e
Which are depth first traversals of the above graph?

(a) I, II, and IV only
(b) I and IV only
(c) II, III, and IV only
(d) I, III, and IV only

Ans. (d) : I, III, and IV only

46. **Given below are two statements**
Statement I: In an undirected graph, number of odd degree vertices is even.
Statement II: In an undirected graph, sum of degrees of all vertices is even.
In light of the above statements, choose the correct answer from the options given below

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

Ans. (a) : Statement 1 theorem :- An undirected graph has an even number of vertices of odd degree.

Proof : Let v_1 be the vertices of even degree and v_2 be the vertices of odd degree in an undirected graph G = (V , E) with m edges.

Then

$$\text{even} \rightarrow 2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v)$$

must be even since deg(v)is even for each $v \in V_1$

This sum must be even because 2m is even and the sum of the degrees of the vertices of even degree is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph there must be an even number of such vertices.

Statement -II :-The degree of any vertex is the number of edges or arcs or lines connected to it. The sum of degrees of any graph can be worked out by adding the degree of each vertex in the graph. The sum of degrees is twice the number of edges. Therefore the sum of degrees is always even.

Theorem Let G = (V, E) be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v)$$

47. Which of the given options provides the increasing order of asymptotic complexity of functions f1, f2, f3 and f4?

A. $f1(n) = 2^n$

B. $f2(n) = n^{3/2}$

C. f3(n) = n log n

D. $f4(n) = n^{\log n}$

Choose the correct answer from the options given below

(a) C, B, D, A

(b) C, B, A, D

(c) B, C, A, D

(d) B, C, D, A

Ans. (a) : $n \log n > n^{3/2} > n^{\log n} > 2^n$

48. The order of a leaf node in a B+ tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

(a) 63

(b) 64

(c) 67

(d) 68

Ans. (a) : Disk block size = 1024 bytes.

Data record pointer size, r = 7 bytes

Value size, V = 9 bytes

Disk Block ptr, P = 6 bytes.

Let order of leaf be m.A leaf node in B + tree contains at most m record pointers, at most m values, and one disk block pointer.

r × m + V × m + P < = 1024

16 m < = 1018

m = < 63

49. A hash function h defined as h(key)=key mod 7, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?

(a) 3 (b) 4

(c) 5 (d) 6

Ans. (c) : Keys 44, 45, 79, 55, 91, 18, 63

h (key) = key mod 7

h (44) = 44 mod 7 = 2

h (45) = 45 mod 7 = 3

h (79) = 79 mod 7 = 2

but 2 is already filled 44, linear probing is applied but 3 is also filled.

So 79 will occupy 4

h (55) = 55 mod 7 = 6

h (91) = 91 mod 7 = 0

h (18) = 18 mod 7 = 4

but 4 is occupied by 79 So, it will occupy 5.

h (63) = 63 mod 7 = 0 , 0 is also occupied so, it will occupy1.

50. In the following table, the left column contains the names of standard graph algorithms and the right column contains the time complexities of the algorithms. Here, *n* and *m* are number of vertices and edges, respectively. Match each algorithm with its time complexity.

	List - I Standard graph algorithms		List - II Time complexities
A.	Bellman - Ford algorithm	I.	O(m*log n)
B.	Kruskal's algorithm	II.	O(n^3)
C.	Floyd - Warshall algorithm	III.	O(n*m)
D.	Topological sorting	IV.	O(n+m)

Choose the correct answer from the options given below :

(a) A - III, B - I, C - II, D - IV

(b) A - II, B - IV, C - III, D - I

(c) A - III, B - IV, C - I, D - II

(d) A - II, B - I, C - III, D - IV

Ans. (a):	List - I Standard graph algorithms		List - II Time complexities
A.	Bellman - Ford algorithm	III.	O(n*m)
B.	Kruskal's algorithm	I.	O(m*log n)
C.	Floyd - Warshall algorithm	II.	$O(n^3)$
D.	Topological sorting	IV.	O(n+m)

51. Which agent deals with the happy and unhappy state?

(a) Utility - based agent

(b) Model - based agent

(c) Goal - based Agent

(d) Learning Agent

Ans. (a) : Utility Based Agent :-Utility based agent choose actions on the basis of utilities or preference and choose the actions that maximize the expected utility. Utility describes the happiness of the agent.

Model Based Agent :- Model based agents have a model i.e. knowledge about how things happen in the whole world & based on the model it performs actions.

Goal based Agent :- Goal based agents expand the capabilities of the model - based agent by having the "goal" information. They choose an action, so that they can achieve the goals. These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not.

Learning agents :- learn from past experiences and automatically adapt through learning. It has learning capabilities.

Hence, Utility - based agents deal with happy & unhappy agents.

52. Given below are two statements

Statement I: Breadth - First Search is optimal when all the step costs are equal whereas uniform - cost search is optimal with any step - cost.

Statement II: When all the step costs are same uniform - cost search expends more nodes at depth d than the Breadth - First Search.

In light of the above statements, choose the correct answer from the options given below

(a) Both Statement I and Statement II are true

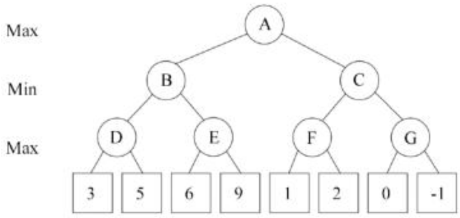
(b) Both Statement I and Statement II are false

(c) Statement I is true but Statement II is false

(d) Statement I is false but Statement II is true

Ans. (a) : Both Statement I and Statement II are true.

53. Consider the given tree below. Calculate the value at the root of the tree using alpha - beta pruning algorithm.



(a) 3
(b) 5
(c) 6
(d) 9

Ans. (b) : 5

54. Which among the following statement(s) is(are) FALSE?

A. Greedy best - first search is not optimal but is often efficient.

B. A* is complete and optimal provided *h(n)* is admissible or consistent.

C. Recursive best - first search is efficient in terms of time complexity but poor in terms of space complexity.

D. *h(n)* = 0 is an admissible heuristic for the 8 - puzzle.

(a) A only
(b) A and D only
(c) C only
(d) C and D only

Ans. (c) : C is false only.

55. Consider the sentence below.

There is a country that borders both India and Pakistan.

Which of the following logical expressions express the above sentence correctly when the predicate Country(x) represents that *x* is a country and Borders(x, y) represents that the countries *x* and *y* share the border?

(a) ∃c Country(c) ^∧ Border (c, India) ∧ Border (c,Pakistan)
(b) ∃c Country(c) ⇒ [Border (c, India) ∧ Border (c,Pakistan)]
(c) [∃c Country(c)] ⇒ [Border (c, India) ∧ Border (c,Pakistan)]
(d) ∃c Border (Country(c), India ∧ Pakistan)

Ans. (a) : ∃c Country(c) ∧ Border (c, India) ∧ Border (c,Pakistan)

56. Next five questions are based on the following passage.

Consider a domain consisting of three Boolean variables Toothache, Cavity, and Catch. The

full joint distribution is a $2 \times 2 \times 2$ table as shown in the figure below.

	toothache		¬toothache	
	catch	¬catch	catch	¬ catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

The marginal probability of cavity P(cavity) is __________.

(a) 0.200
(b) 0.216
(c) 0.120
(d) 0.080

Ans. (a) : 0.200
Probability (Cavity)
= 0.108 + 0.012 + 0.072 + 0.008
= 0.200

57. The probability of a cavity, given evidence of a toothache, P(cavity | toothache) is __________.

(a) 0.400
(b) 0.600
(c) 0.280
(d) 0.216

Ans. (b) :

$$= P(\text{Cavity} \mid \text{toothache}) = \frac{P(\text{Cavity} \cap \text{toothache})}{P(\text{toothache})}$$

$$= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064}$$

$$= \frac{0.120}{0.200} = 0.600$$

58. The probability of a toothache, given evidence of a cavity, P(toothache | cavity) is __________.

(a) 0.400
(b) 0.600
(c) 0.280
(d) 0.216

Ans. (b) : P(toothache | cavity)
= P(toothache ∧cavity)|P(cavity)
= (0.108 + 0.012)|(0.108 + 0.012 + 0.072 + 0.008)
= 0.600

59. P(cavity ∨ toothache) is __________.

(a) 0.200
(b) 0.120
(c) 0.280
(d) 0.600

Ans. (c) :
= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064
= 0.280

60. The probability for Cavity, given that either Toothache or Catch is true, P(Cavity | toothache ∨ catch) is _______.

(a) 0.6000

(b) 0.5384

(c) 0.8000

(d) 0.4615

Ans. (d) : P(cavity | toothache ∨ catch)

= cavity and (toothache or catch) | (catch or toothache)

= (0.108 + 0.012 + 0.072)| 0.416

= 0.4615

61. Which of the DBMS component ensures that concurrent execution of multiple operations on the database results into a consistent database state?

(a) Logs

(b) Buffer manager

(c) File manager

(d) Transaction processing system

Ans. (d) :Transaction Processing System :- A transaction processing system (TPS) is a type of information system that collect, stores, modifies and retrieves the data transactions of an enterprise.

In order to qualify as a TPS, transactions made by the system must pass the ACID test. The ACID tests refers to the following four requisites : -

1) Atomicity

2) Consistency

3) Isolation

4) Durability

TPS provides, Rapid response, continuous availability data integrity, Ease of use.

ex - Online transaction processing (OLTP), Real time transaction.

62. Consider following two statements:

Statement I: Relational database schema represents the logical design of the database.

Statement II: Current snapshot of a relation only provides the degree of the relation.

In the context to the above statements, choose the correct option from the options given below:

(a) Statement I is TRUE but Statement II is FALSE

(b) Statement I is FALSE but Statement II is TRUE

(c) Both Statement I and Statement II are FALSE

(d) Both Statement I and Statement II are TRUE

Ans. (a) : Degree of relationship represents the number of entity types that associated in a relationship. Types of degree are, unary, Binary, Temary, N- ary.

63. Given a fixed-length record file that is ordered on the key field. The file needs B disk blocks to store R number of records. Find the average access time needed to access any record of the given file using binary search.

(a) $\frac{B}{2}$

(b) B + R

(c) B

(d) $\log_2 B$

Ans. (d) : $\log_2 B$

64. Given the following STUDENT-COURSE scheme: STUDENT (Rollno, Name, courseno) COURSE (courseno, coursename, capacity), where Rollno is the primary key of relation STUDENT and courseno is the primary key of relation COURSE. Attribute coursename of COURSE takes unique values only. Which of the following query(ies) will find total number of students enrolled in each course, along with its coursename.

A. SELECT coursename, count(*) 'total' from STUDENT natural join COURSE group by coursename;

B. SELECT C.coursename, count(*) 'total' from STUDENT S, COURSE C where S.courseno=C.courseno group by coursename;

C. SELECT coursename, count(*) 'total' from COURSE C where courseno in (SELECT courseno from STUDENT);

(a) A and B only

(b) C only

(c) A only

(d) B only

Ans. (a) : A and B only

65. Given the following STUDENT-COURSE scheme: STUDENT (Rollno, Name, Courseno) COURSE (Courseno, Coursename, Capacity), where Rollno is the primary key of relation STUDENT and Courseno is the primary key of relation COURSE. Attribute Coursename of COURSE takes unique values only. The number of records in COURSE and STUDENT tables are 3 and 5 respectively. Following relational algebra query is executed: R=STUDENT X COURSE

Match List-I with List-II in context to the above problem statement.

	List - I		List - II
A.	**Degree of table R**	**i.**	**15**
B.	**Cardinality of table R**	**ii.**	**NIL**

C.	**Foreign key of relation STUDENT**	**iii.**	**6**
D.	**Foreign key of relation COURSE**	**iv.**	**Courseno**

Choose the correct answer from the options given below:

(a) A - III , B - I , C - IV , D - II

(b) A - I , B - III , C - IV , D - II

(c) A - I , B - III , C - IV , D - II

(d) A - I , B - III, C - II , D - IV

Ans. (a) :

	List -I		**List-II**
A.	Degree of table R	iii.	6
B.	Cardinality of table R	i.	15
C.	Foreign key of relation STUDENT	iv.	Courseno
D.	Foreign key of relation COURSE	ii.	NIL

66. Suppose a B^+ tree is used for indexing a database file. Consider the following information: size of the search key field= 10 bytes, block size = 1024 bytes, size of the record pointer= 9 bytes, size of the block pointer= 8 bytes.

Let K be the order of internal node and L be the order of leaf node of B^+ tree, then (K, L)=______.

(a) (57, 53)

(b) (50, 52)

(c) (60, 64)

(d) (34, 31)

Ans. (a) : 57, 53

Sol. **Internal Node :-**

$n * P_b + (n - 1)\text{keys} \le BS$

$k * 8 + (k - 1)\ 10 \le 1024$

$8k + 10k - 10 \le 1024$

$k \le \frac{1024}{18} = 57$

Leaf Node :-

$P_b\ (n - 1)\ (\text{key} + \text{record}) \le BS$

$8 + (L - 1)(10 + 9) \le 1024$

$19L \le 1033$

$L \le = \frac{1033}{19} = 53$

67. Given a relation scheme R(x,y,z,w) with functional dependencies set F={x y, z w}. All attributes take single and atomic values only.

A. Relation R is in First Normal FORM

B. Relation R is in Second Normal FORM

C. Primary key of R is xz

Choose the correct answer from the options given below:

(a) C only
(b) B and C only
(c) A and C only
(d) B only

Ans. (c) : A and C only

68. A company is consuming parts in the manufacturing of other products. Each of the part is either manufactured within the company or purchased from the external suppliers or both. For each part, partnumber, partname is maintained. Attribute batchnumber is maintained if the consumed part is manufactured in the company. If part is purchased from external supplier, then supplier name is maintained. Which of the following constraints need to be considered when modelling class/subclass concepts in ERD for the given problem.

(a) Total specialization and overlapping constraints
(b) Disjoint constraint only
(c) Partial participation
(d) Partial participation and disjoint constraints

Ans. (a) : Total specialization and overlapping constraints

69. A transaction may be in one of the following states during its execution life cycle in concurrent execution environment.

A. FAILED
B. TERMINATED
C. PARTIALLY COMMITTED
D. COMMITTED
E. ACTIVE

Given a transaction in active state during its execution, find its next transitioned state from the options given below:

(a) A only
(b) Either A or C only
(c) C only
(d) D only

Ans. (b) : Either A or C only

70. Which of the following is used to create a database schema?

(a) DML
(b) DDL
(c) HTML
(d) XML

Ans. (b) : The DDL commands in SQL are used to create database schema and to define the type and structure of the data that will be stored in a data base. Sql DDL commands are further divided into the following major categories : create, alter, drop, truncate.

71. Given memory access time as p nanoseconds and additional q nanoseconds for handling the page fault. What is the effective memory access time if a page fault occurs once for every 100 instructions?

(a) $p+\frac{q}{100}$

(b) $\frac{p+q}{100}$

(c) p + q

(d) $\frac{p}{100}+q$

Ans. (a) : Given, memory Access Time = P nanoseconds.

Page Fault service Time = q nanoseconds.

one page fault occurs for every 100 instructions.

Formula:

Effective Memory Access Time (EMAT) =

Page fault rate * Page fault service time + (1 – Page fault rate) * Memory Access time

Calculation:

Page fault rate = 1/100 {since, 1 page every 100 instruction}

EMAT : 1/100 * q + (1 – 1/100) * P

$$=\frac{q}{100}+P-\frac{P}{100}$$

$$=P+\frac{q}{100}$$ {Since, $\frac{P}{100}$ is too small, we can neglect it}

72. In a file allocation system, the following allocation schemes are used:

A. Contiguous

B. Indexed

C. Linked allocation

Which of the allocation scheme(s) given above will not suffer from external fragmentation? Choose the *correct* answer from the options given below:

(a) A only

(b) B and C only

(c) A and B only

(d) C only

Ans. (b) : Contiguous allocation suffer from external fragmentation. But Linked and indexed allocations are non-contiguous, so they will not suffer from **external fragmentation**.

73. Given below are three statements related to interrupt handling mechanism

A. Interrupt handler routine is not stored at a fixed address in the memory.

B. CPU hardware has a dedicated wire called the interrupt request line used for handling interrupts

C. Interrupt vector contains the memory addresses for speciaized interrupt handlers.

In the context of above statements, choose the correct answer from the options given below:

(a) A is TRUE only
(b) Both B and C are TRUE only
(c) Both A and B are TRUE only
(d) Both A, C are TRUE only

Ans. (b) : Both B and C are TRUE only

74. Match List-I with List-II

	List - I System calls		List - II Description
A.	fork()	i.	Sends a signal from one process to another process
B.	exec()	ii.	Indicates termination of the current process
C.	kill()	iii.	Loads the specified program in the memory
D.	exit()	iv.	Creates a child process

Choose the correct answer from the options given below:

(a) A - II , B - III , C - IV , D - I
(b) A - IV , B - III , C - I , D - II
(c) A - IV , B - I , C - II , D - III
(d) A - IV , B - III , C - II, D - I

Ans. (b) : Fork () : Fork system call is used for creating a new process, which is called child process, which runs currently with the process that makes the fork() call (pavement process.)

exec() : The exec() family of functions replaces the current process image with a new process image. It loads the program into the current process space and runs it from the entry point.

Kill() : kill is a command that is used in several, popular operating systems to send signals to running process.

Exit() : exit() terminates the calling process "immediately"

75. Consider the following 3 processes with the length of the CPU burst time given in milliseconds:

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9

What is the average waiting time for these processes if they are scheduled using preemptive shortest job first scheduling algorithm?

(a) 5.5 (b) 2.66
(c) 4.66 (d) 6

Ans.(c): Whenever new process arrives, there may be pre-emption of the running process, based on burst time.

Process	Arrival time	Burst Time	completion CT	TAT	WT	Response Time
P1	0	8	12	12	4	0
P2	1	4	5	4	0	0
P3	2	9	21	19	10	11
				$Avg = \frac{35}{5}$ $= 11.33$	$Avg = \frac{14}{3}$ $= 4.66$	

P1	P2	P2	P1	P3

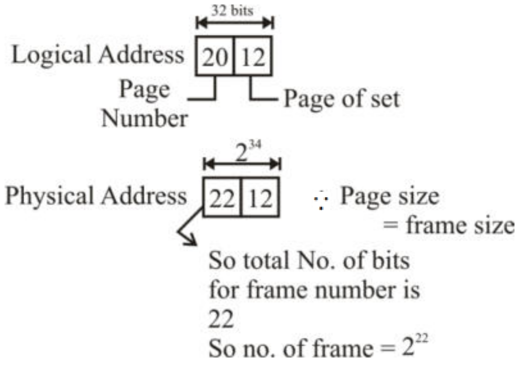
0 1 2 5 12 21

76. Read the following and answer the questions:
Consider a machine with 16 GB main memory and 32 - bits virtual address space, with page size as 4KB. Frame size and page size is same for the given machine.
The number of bits reserved for the frame offset is ______.

(a) 12
(b) 14
(c) 32
(d) 8

Ans. (a) : Physical Address space = 16 GB
$= 2^4 \times 2^{30} = 2^{34}$

Virtual Address space = 2^{32}
Page size = 4 kb = $2^2 \times 2^{10} = 2^{12}$



So bits reserved for frame of set = 12

77. Read the following and answer the questions:
Consider a machine with 16 GB main memory and 32 - bits virtual address space, with page size as 4KB. Frame size and page size is same for the given machine.
Find number of pages required for the given virtual address space.

(a) 2^{10}
(b) 2^{20}
(c) 2^{30}
(d) 2^{12}

Ans. (b) : 4 kb = $2^2 \times 2^{10} = 2^{12}$ B

No. of page = $= \dfrac{\text{LAS}}{\text{page size}} = \dfrac{2^{32}}{2^{12}}$

$= 2^{32-12} = 2^{20}$

78. Read the following and answer the questions: Consider a machine with 16 GB main memory and 32 - bits virtual address space, with page size as 4KB. Frame size and page size is same for the given machine.
What is the minimum number of bits needed for the physical address?

(a) 28
(b) 34
(c) 24
(d) 12

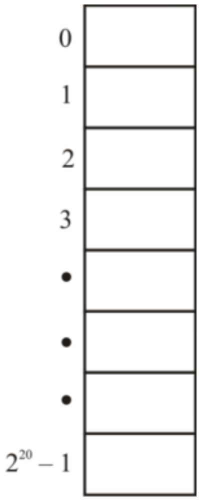
Ans. (b) : ∵ Physical Address space = 16 GB = 2^{34}
∴ 34 number of bits needed for the physical address.

79. Read the following and answer the questions: Consider a machine with 16 GB main memory and 32 - bits virtual address space, with page size as 4KB. Frame size and page size is same for the given machine.
What is the size of page table for handling the given virtual address space, given that each page table entry is of size 2 bytes?

(a) 2MB
(b) 2 KB
(c) 32 MB
(d) 12KB

Ans. (a) : No. of entries in page table = No. of page in a process = 2^{20}

Size of page table = 2×2^{20}
$= 2 \times 1$ MB
$= 2$ MB



80. Read the following and answer the questions: Consider a machine with 16 GB main memory and 32 - bits virtual address space, with page

size as 4KB. Frame size and page size is same for the given machine.

If a process of size 34KB is to be executed on this machine, then what will be the size of internal fragmentation for this process?

(a) 4 KB
(b) Zero
(c) 1KB
(d) 2 KB

Ans. (d) : 2 KB

Page size = 4 KB

= Break process into pages

$= \frac{34KB}{4KB} = 8.---$

alot Fragmetion 8 time

4 KB	4 KB	4 KB	------- 8 time

34 – 4 = 30	8×4 = 32 KB
30 – 4 = 26	process of size = 34 KB
26 – 4 = 22	34 – 32 = 2 KB
... ...	(Freagmentaion) **Ans.**
... ...	
8 time	

⇒

81. Match List-I with List-II

	List -I		List -II
A.	Odd Function	i.	NAND gate
B.	Universal Gate	ii.	XOR gate
C.	2421 code	iii.	Amplification
D.	Buffer	iv.	Self - Complementing

Choose the correct answer from the options given below:

(a) A - I, B - II, C - III, D - IV
(b) A - II, B - I, C - IV, D - III
(c) A - I, B - III, C - II, D - IV
(d) A - IV, B - II, C - III, D - I

Ans. (b) : A - II, B - I, C - IV, D - III

82. The Octal equivalent of hexadecimal $(D.C)_{16}$ is:

(a) $(15.6)_8$
(b) $(61.6)_8$
(c) $(15.3)_8$
(d) $(61.3)_8$

Ans. (a) : $(15.6)_8$

83. A digital computer has a common bus system for 8 registers 16 bits each. How many multiplexers are required to implement common bus? What size of multiplexers is required?

(a) 16, 8x1
(b) 8, 16x1
(c) 8, 8x1
(d) 16, 16x1

Ans. (a) : N = 8, K = 16
If there are 'N' number of K-bit registers
then,
No. of mux = K = 16
Size of mux = N×1 = 8×1

84. Which of the following is not an example of pseudo - instruction?
(a) ORG
(b) DEC
(c) END
(d) HLT

Ans. (d) : HLT
Pseudo - instruction :-
⇒ do not perform any operation on data
⇒ giving some into to the a sembler
⇒ 4 Pseudo - instn,
ORG → origin
END
Dec → decimal
HEX

85. The reverse Polish notation of the following infix expression [A*{B+C*(D+E)}] / {F*(G+H)} is__________.
(a) ABCDE+*+*FGH+*/
(b) ABCDE*++*FGH+*/
(c) ABCDE+*+*FGH*+/
(d) ABCDE+**+FGH+*/

Ans. (a) : ABCDE+*+*FGH+*/

86. Arrange the following in the increasing order of complexity.
A. I/O Module
B. I/O processor
C. I/O Channel
D. DMA
Choose the correct answer from the options given below
(a) D, C, B, A
(b) C, D, A, B
(c) A, B, C, D
(d) A, D, C, B

Ans. (d) : I/O Module<DMA< I/O channel< I/O processar.

87. The characteristics of the combinational circuits are:

A. Output at any time is function of inputs at that time

B. Contains memory elements

C. Do not have feedback paths

D. Clock is used to trigger the circuits to obtain outputs

Choose the correct answer from the options given below:

(a) A and B only
(b) B and C only
(c) A and C only
(d) B and D only

Ans. (c) : A and C only

88. The cache coherence problem can be solved

A. by having multiport memory

B. allow only nonshared data to be stored in cache

C. using a snoopy cache controller

D. uisng memory interleaving

Choose the correct answer from the options given below:

(a) A and C only
(b) B and C only
(c) D and C only
(d) B and D only

Ans. (b) : B and C only

89. Given below are two statements

Statement I: CISC computers have a large of number of addressing modes.

Statement II: In RISC machines memory access is limited to load and store instructions.

In light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

Ans. (a) : Both statements are true

90. Which of the following statement is true?

A. Control memory is part of the hardwired control unit.

B. Program control instructions are used to alter the sequential flow of the program.

C. The register indirect addressing mode for accessing memory operand is similar to displacement addressing mode.

D. CPU utilization is not affected by the introduction of Interrupts.

(a) A
(b) B
(c) C
(d) D

Ans. (b) : Program Control instruction :-The program control instructions control the flow of program execution and are capable of branching to different program segments. Some of the program control instructions are listed in the table.

Name	**Mnemonics**
Branch	BR
Jump	JMP
Skip	SKP
Call	Call
Return	RET
Compare (by subtraction)	CMP
Test	TST

91. Match List-I with List-II

	List - I		**List - II**
A.	**Data Link Layer**	**i.**	**True end-to-end layer**
B.	**Network Layer**	**ii.**	**Token Management**
C.	**Transport Layer**	**iii.**	**Produce billing information**
D.	**Session Layer**	**iv.**	**Piggybacking**

Choose the correct answer from the options given below:

(a) A - IV, B - III, C - I, D - II
(b) A - IV, B - II, C - I, D - III
(c) A - II, B - III, C - I, D - IV
(d) A - IV, B - I, C - III, D - II

Ans. (a) : Data Link Layer : pigybacking is sometimes referred to as "wi-fi squatting". The usual purpose of piggybacking is simply to gain free network access rather than any malicious intent, but it can slow down data transfer for legitimate users of network.

Transport Layer : Responsible for true end-to-end layer from source to destination. It is termed as and end-to-end layer because it provides a point to point connection rather than hop to hop. It reliably deliver the services between the source-host and destination host.

Session Layer : is responsible for establishing managing, synchronizing and termination session between end-user application processes. It works as a dialog, controller. It allows the system to communicate in either. Half duplex or full duplex mode of communication. It is responsible for token management.

Network Layer :- subnet usage accounting : has accounting functions to keep track of frames forwarded by subnet intermediate systems, to produce billing information.

92. Given below are two statements

Statement I : Telnet, Ftp, Http are application layer protocol

Statement II: The Iridium project was planned to launch 66 low orbit satellites.

In light of the above statements, choose the correct answer from the options given below:

(a) Both Statement I and Statement II are true

(b) Both Statement I and Statement II are false

(c) Statement I is true but Statement II is false

(d) Statement I is false but Statement II is true

Ans. (a) : Both statement are true.

93. Which of the following statement is False?

(a) Packet switching leads to better utilization of bandwidth resources than circuit switching.

(b) Packet switching results in less variation in the delay than circuit switching.

(c) Packet switching sender and receiver can use any bit rate, format or framing method unlike circuit switching.

(d) Packet switching can lead to reordering unlike circuit switching

Ans. (b & c) : Packet– Switched networks move data in separate, small packets based on the destination address in each packet. When received, packets are reassembled in the proper sequence to make up the message.

Circuit- Switched networks : required dedicated point-to-point connections during calls.

- Packet switching leads to better utilization of bandwidth resources than circuit switching // Because data move in packets therefore better use of resources.
- Packet switching results in less variation in delay than circuit switching // Queueing delay flunctuation is more in packet switching as there is no dedicated path.
- Packet switching requires more per packet processing than circuit switching // All the info needs to be in packet as well as packet reassembling has to be done at end therefore more processing.
- Packet switching can lead to reordering unlike in circuit switching // because reassembling is done at destination, packing reordering might occur.

94. In Ethernet when Manchester coding is used, the bit rate is__________.

(a) Half the baud rate
(b) twice the baud rate
(c) Thrice the baud rate
(d) Same as the baud rate

Ans. (a) : In manchester encoding, 2 signals changes to represent a bit. So the baud rate = 2*bit rate so that bit rate is half the baud rate.

95. The address of class B host is to be split into subnets with 6 - bit subnet number. What is the maximum number of the subnets and the maximum number of hosts in each subnet?

(a) 62 subnets and 262142 hosts
(b) 64 subnets and 1024 hosts
(c) 64 subnets and hosts 262142
(d) 62 subnets and 1022 hosts

Ans. (d) : In class B
First 2 octet are reserved for NID and remaining for HID... so first 6 - bits of 3rd octet are used for subnet and remaining 10 bits for host... maximum number of subnets = $2^6 - 2 = 62$ Note that 2 is subtracted because subnet value consisting of all zeros and all ones (broadcast), reducing the number of available subnets by two in classical subnetting. In modern networks we can have 64 as well and no of hosts = $2^{10} - 2 = 1022$ so total 62 subnets and 1022 hosts.

96. In electronic mail, which of the following protocols allows the transfer of multimedia?

(a) IMAP
(b) SMTP
(c) POP3
(d) MIME

Ans. (d) :IMAP :- is for the retrieval of emails and SMTP is for the sending of emails. That means IMAP talks to both the client and server to get emails, and SMTP talks only to servers to send emails.

POP3/IMAP :- Post office protocol (POP) is an application layer Internet standard protocol used by local e-mail clients to retrieve email from a remote server over a TCP/IP connection. Virtually all modern e-mail clients and servers support POP3, and it along with IMAP (Internet Message Access Protocols) are the two most prevalent Internet standard protocols for email retrieval, with IMAP (Internet Message Access Protocol) are the two most prevalent Internet standard protocols for emails retrieval, with many webmail service provides such as Google mail, Microsoft mail and Yahoo!

MIME :- MIME stands for multipurpose Internet mail Extensions. It is used to extend the capabilities of Internet e-mail protocols such as SMTP. The MIME protocols allows the users, to exchange various types of digital content such as pictures, audio, video, and various types of documents and files in the e-mails.

97. A message is encrypted using public key cryptography to send a message from sender to receiver. Which one of the following statements is True?

(a) Sender encrypts using receiver's public key

(b) Sender encrypts using his own public key

(c) Receiver decrypts using sender's public key

(d) Receiver decrypts using his own public key

Ans. (a) : In Public key cryptography, both sender and receiver generate a pair of keys, – public key and private key. Public keys are known globally. suppose A is sender and B is the receiver.

So, A has 3 keys :

1. Public key of A (everyone knows)
2. Private key of A (only A knows)
3. Public key of B (Everyone knows)

And B also has 3 keys :

1. Public key of B (Everyone knows)
2. Private key of B (only B knows)
3. Public key of A (Everyone knows)

• Anything that is encrypted using public key of A can be decrypted only using private key of A.

• Anything that is encrypted using private key of A can be decrypted only using public key of A.

• Anything that is encrypted using public key of B can be decrypted only using private key of B.

• Anything that is encrypted using private key of B can be decrypted only using public key of B.

Now A wants to send a secret message to B. so for encryption : A has following 3 options :

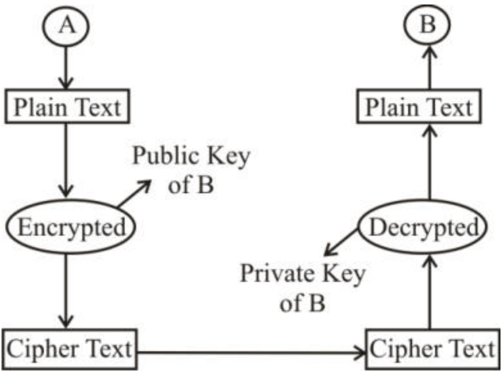
1- Public key of A (Everyone knows) : So for decryption B needs - Private key of A - only A knows it. So, B will not be able to decrypt it.

2. Private key of A (only A knows) So, for decryption B needs - public key of A - Everyone known it. So everyone can decrypt it. So it is of no use.

3. Public key of B (Everyone knows) : So, for decryption B needs - private key of B - only B knows it. So only B will able to decrypt it (That's what we want)

So for providing security.

Sender encrypts using receiver's public key and Receiver decrypts using his own.



So, connect option is (A) sender encrypts using receiver's public key.

98. Which of the following statements are true?

A. Frequency division multiplexing technique can be handled by digital circuits.

B. Time division multiplexing technique can be handled by analog circuits

C. Wavelength division multiplexing technique is used with optical fiber for combining two signals.

D. Frequency division multiplexing technique can be applied when the bandwidth of a link is greater than the bandwidth of the signals to be transmitted.

Choose the correct answer from the options given below:

(a) B and D only (b) C and D only

(c) A and D only (d) B and D only

Ans. (b) : C and D only

99. Which of the following statements are true?

A. X.25 is connection-oriented network

B. X.25 doesn't support switched virtual circuits.

C. Frame relay service provides acknowledgements.

D. Frame relay service provides detection of transmission errors.

Choose the correct answer from the options given below:

(a) A and D only (b) B and D only

(c) C and D only (d) B and C only

Ans. (a) : A and D only

100. Let G(x) be the generator polynomial used for CRC checking. The condition that should be satisfied by the G(x) to catch all errors consisting of an odd number of inverted bits is:

(a) $(x+1)$ is factor of G(x)

(b) $(x - 1)$ is factor of G(x)

(c) $(x^2 +1)$ is factor of G(x)

(d) $(1 - x^2)$ is factor of G(x)

Ans. (a) : $(x+1)$ is factor of G(x)